Who Is Really on the Frontier?
Confidence Sets for Cost-Accuracy Pareto Leaderboards

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Abstract

A leaderboard that reports cost beside accuracy is read through its Pareto frontier, the systems that no cheaper system beats. The frontier is drawn from point estimates on a sample of tasks, so it is an estimate, and no leaderboard reports how uncertain it is. We give a simultaneous confidence band for the frontier curve and a confidence set for its members that contains the whole true frontier with probability at least $1 - \alpha$. The band is proved for fixed or per-task costs. The set is proved for fixed costs, and for per-task costs when the true mean costs are distinct; a tie-robust set drops that condition at the price of a larger set. We check both on known truth in 15 scenarios and report where they fail, including nearly tied costs and cheaper near-copies of frontier systems. We rebuild per-task costs for 356 runs on 11 benchmarks of the Holistic Agent Leaderboard (HAL) from its public call logs. HAL's cost totals double count nested trace spans on 202 of 332 runs. Counting each call once changes HAL's frontier on 2 benchmarks, and counting and pricing together change it on 8 of 11. With uncertainty added, the share of systems that cannot be excluded from the frontier runs from 42% to 100% across boards, and excluding half of the systems that cannot now be excluded would take 122 to 7,863 tasks, depending on the board. Certificates for new frontier entrants and a winner's-curse correction for frontier members are in the appendices, each stated under its own null. The methods are in the open-source package `honestboard`.

1 Introduction

Leaderboards for AI agents increasingly report what a run costs beside how often it succeeds. The Holistic Agent Leaderboard (HAL) [15] does this for every benchmark it hosts, and it draws the cost-accuracy picture the way an economist would: a system is worth looking at only if no cheaper system does at least as well. The systems that pass this test form the Pareto frontier, and HAL's own write-up stresses how few systems sit on it. Readers and the teams that build agents act on that picture. A system on the frontier is a reasonable one to consider at its price, and a system below it usually is not.

The picture is drawn from point estimates. Each system's accuracy is its success rate on a sample of tasks: 33 on AssistantBench, 45 on CORE-Bench hard, 50 on tau-bench airline and on the SWE-bench Verified mini split. With samples that small the rate carries a standard error of several points, and the frontier, which is a function of those rates, moves when the rates move.

A system just below the frontier might sit on it under another draw of tasks, and a system on it might not. HAL reports no uncertainty for its frontiers.

This paper supplies that uncertainty. It answers four questions.

1. How high can the frontier curve, the best accuracy available at each cost, really be, and how low; and which systems cannot be excluded from the frontier at a stated confidence level?
2. How wrong is the plain frontier when the truth is known, and where do the guarantees fail?
3. What is the cheapest system a reader cannot yet rule out as the best?
4. How many tasks would a board need before its frontier is resolved?

It also audits HAL with per-task costs rebuilt from its call logs, and it gives, in two appendices, certificates for new frontier entrants and a winner's-curse correction for frontier members.

Contributions.

- A simultaneous confidence band for the frontier curve itself, valid with fixed or per-task costs and with no condition on ties among the true mean costs (Section 3.2).
- A confidence set for the frontier's members that contains the whole true frontier with probability at least $1 - \alpha$, proved for fixed costs and for per-task costs without ties; a tie-robust version that drops the condition on ties; and exact finite-sample sets (Section 3.3).
- Resolution numbers, the task count at which a board would start to resolve its own frontier, and two descriptive summaries, a bootstrap frontier stability and the cheapest unresolved competitor of the best (Sections 3.3 and 3.4).
- A known-truth validation in 15 scenarios, including near-copy systems, outcome-dependent costs, nearly tied costs and clustered tasks, a stability check on held-out halves of real boards, and an account of where the guarantees fail (Section 5).
- An audit of HAL with consistent costing: each model call counted once, one price table, the double count quantified for every run, the effects of counting and of pricing on HAL's published frontiers reported separately, and the sensitivity of the audit to the costing rules (Sections 4 and 6).
- In the appendices, two certificates for a new frontier entrant with chain-wise error control, one a record claim and one a certificate of Pareto membership under a stated null, a proof that no certificate built on the entrant's strongest rival alone can hold its level, and the winner's-curse correction for frontier members.

What the paper does not claim. The tools are adapted, not invented from nothing. The set is Hsu's multiple-comparisons-with-the-best construction with each system's rivals restricted to the systems that cost no more and the margin taken jointly, the exact sets are paired sign tests with Bonferroni, and the certificates are the selective tests of a companion paper with a different conditioning event. The inference is conditional on the submitted systems and takes the tasks as draws from the benchmark's task process. It does not account for tuning on public tasks or for which systems were submitted (Section 2).

Related work. Uncertainty in leaderboard rankings has been treated through confidence sets for ranks [21–23], confidence sets of the best [12, 13], the winner's curse of a selected system [1, 4], and standard errors for language-model evaluations [8, 19, 20]. The observed frontier is a free-disposal-hull estimator, and inference for such frontiers is developed in econometrics for sampled firms with exactly measured inputs and outputs: the asymptotics of the free-disposal-hull estimator [24] and the bootstrap for nonparametric frontier models [25, 26]. Tests of stochastic dominance [17] ask a different question. The sampling error of an efficient frontier is studied in finance [9] and that of Pareto fronts under noisy objectives in evolutionary optimisation [11]. Here the units are systems whose coordinates, accuracy and cost, are both estimated from the same tasks, and the tasks are shared across units. Identifying a Pareto set from noisy observations has also been studied where the analyst chooses what to sample next: Pareto-set identification with (ϵ, δ) guarantees from bandit feedback [3], active learning of the Pareto front [29], and multi-objective ranking and selection in simulation [2]. Cost has entered leaderboard design through HAL [15], the argument that agent evaluations should report it [14], and the cost-of-pass framework [10], all from point estimates. To our knowledge no earlier work gives a published cost-accuracy frontier a simultaneous confidence band and a confidence set for its members, though the literatures above are close, and a systematic comparison with them is beyond one paper.

Findings. HAL's cost totals double count nested trace spans on 202 of 332 runs and carry the prices of the month each run was made. Counting each call once changes the point-estimate frontier on 2 benchmarks; counting and pricing together change it on 8 of 11, with 10 of the 56 systems on the frontiers drawn from HAL's reported totals leaving and 18 others joining, 28 changes in all, 24 of those changes through the pricing and 4 through the counting (Section 6). With uncertainty added, 64 systems sit on the observed frontiers, but the share that cannot be excluded runs from 42% on USACO to 100% on AssistantBench and SciCode, and tends to fall as the task count grows, though not monotonically. In the base simulations the set keeps the whole true frontier in 0.95 to 0.98 of boards at the nominal 0.95. With nearly tied true costs the standard set keeps it in only 0.26, and with cheaper near-copies of frontier systems in 0.80. The tie-robust set keeps it in every board, in the nearly tied case only by returning the whole board, because the data cannot resolve such a frontier (Sections 5 and 3.3).

2 Setting

A board has K systems evaluated on the same T tasks. System k scores $y_{kt} \in [0, 1]$ on task t: a pass or fail on most benchmarks, a partial score on some. Its accuracy is the mean $\bar{y}_k = T^{-1} \sum_t y_{kt}$, which estimates $\mu_k = \mathbb{E}\, y_{kt}$. Its cost is either one number per system, c_k (the price of the run divided by T), or one number per task, c_{kt}, with $\bar{c}_k$ the mean estimating $\gamma_k = \mathbb{E}\, c_{kt}$; on HAL the per-task costs come from the call logs (Section 4).

System j *dominates* system i when $c_j \le c_i$ and $\bar{y}_j \ge \bar{y}_i$ with at least one of the two strict. The *observed frontier* is the set of systems that no other system dominates. Two systems equal in both cost and accuracy dominate neither the other, so both can be on the frontier; this is the only tie convention used, and it is applied throughout. The *true frontier* is the same set with μ_k and γ_k in place of $\bar{y}_k$ and $\bar{c}_k$: a system is truly undominated when no other system costs no more and is at least as accurate with one of the two strict.

What is estimated. The systems are taken as given. The inference is conditional on the K submitted systems and says nothing about systems that were tried and not submitted, or tuned on

the public tasks. The tasks are draws from the benchmark's task process, independent across tasks and dependent across systems on the same task (hard tasks are hard for everyone). The target is each system's expected accuracy μ_k on new tasks from that process. For a fixed set of tasks the empirical frontier is exactly determined, and the uncertainty described here arises only because the tasks are a sample. The target is a benchmark population, not a deployment population: a result about new kinds of tasks, other repositories or sites, or real use is not covered, and clustered tasks are where it is most at risk (Section 7). Cost is the priced token usage of reproducing the run, in dollars per task under one price table, and its target is the expected value γ_k under the task process. A frontier therefore says which systems are cheapest to reproduce at a given accuracy at the prices of the table, not which were the best buys when they were run. The price table is a fixed list, and a system's cost is the priced token count of its run, which varies with the tasks drawn, so costs are random where prices are not. Three things are not modelled: tuning of a system on the public tasks, run-to-run variation of a system beyond independent task-level noise, and changes in prices over time.

3 Methods

3.1 The observed frontier and why it is not enough

The observed frontier is what every cost-aware leaderboard draws. It is a function of K estimated accuracies, each with a standard error of order $T^{-1/2}$, and it is a discontinuous function: moving one accuracy by a hair can add or remove a system. Section 5 measures how often it is wrong when the truth is known. The short version is that with 300 simulated boards of 20 systems on 50 tasks, 19% of the systems on the observed frontier are truly dominated and 28% of the true frontier is missed.

3.2 A confidence band for the frontier curve

What a cost-aware leaderboard draws is a curve: at each cost x, the best accuracy available from any system whose cost is at most x,

$$A(x) = \max_{j:\,\gamma_j \le x} \mu_j,$$

with γ_j the true mean cost of system j: a staircase that steps up at the cost of each frontier member. The plain frontier is this curve drawn with the sample means $\bar{y}_j$ and $\bar{c}_j$ in place of μ_j and γ_j. A simultaneous band for A follows from a simultaneous band for all $2K$ means at once. Let se_j and sg_j be the standard errors of $\bar{y}_j$ and $\bar{c}_j$ (zero for a cost that is a fixed number), and let c_{band} be the $(1-\alpha)$ quantile, over task resamples that carry outcomes and costs together, of the largest of $|\bar{y}_j^{(b)} - \bar{y}_j|/\mathrm{se}_j$ and $|\bar{c}_j^{(b)} - \bar{c}_j|/\mathrm{sg}_j$ over all systems. Each system then has an accuracy interval $[\bar{y}_j \mp c_{\mathrm{band}}\mathrm{se}_j]$ and a cost interval $[\bar{c}_j \mp c_{\mathrm{band}}\mathrm{sg}_j]$, and

$$L(x) = \max_{j:\ \bar{c}_j + c_{\mathrm{band}}\mathrm{sg}_j \le x} (\bar{y}_j - c_{\mathrm{band}}\mathrm{se}_j), \qquad U(x) = \max_{j:\ \bar{c}_j - c_{\mathrm{band}}\mathrm{sg}_j \le x} (\bar{y}_j + c_{\mathrm{band}}\mathrm{se}_j).$$

A system enters the lower edge only when its cost is surely within the budget x and the upper edge when it possibly is, so neither edge relies on having the cost order right.

Proposition 1 (band)**.** *Let the task vectors of outcomes and costs be i.i.d., with every random accuracy and cost having positive variance, let K be fixed and each true accuracy μ_j fixed in $(0,1)$ as $T \to \infty$. Then*

$$\mathbb{P}\big(L(x) \le A(x) \le U(x) \text{ for every } x \ge \min_j \gamma_j\big) \ \ge\ 1 - \alpha - o(1),$$

with costs fixed or measured per task, and with no condition on ties among the true mean costs. ($A(x)$ is a maximum over an empty set below the cheapest true cost.)

Proof. Let E be the event that every μ_j lies in its accuracy interval and every γ_j in its cost interval. If $\bar{c}_j + c_{\text{band}}\text{sg}_j \le x$ then on E $\gamma_j \le x$, so $\bar{y}_j - c_{\text{band}}\text{se}_j \le \mu_j \le A(x)$; hence $L(x) \le A(x)$. If $\gamma_j \le x$ then on E $\bar{c}_j - c_{\text{band}}\text{sg}_j \le \gamma_j \le x$, so the maximizer j of μ_j over $\{\gamma_j \le x\}$ enters $U(x)$ with $\bar{y}_j + c_{\text{band}}\text{se}_j \ge \mu_j = A(x)$; hence $U(x) \ge A(x)$. Both hold for every x on E. By the multivariate central limit theorem and the consistency of the bootstrap for the mean vector [7], with the standard errors converging in probability by Slutsky's theorem, the bootstrap law of the maximal studentised deviation converges to that of the maximum of the absolute values of a Gaussian vector whose coordinates have positive variance, whose distribution function is continuous and strictly increasing, so $\mathbb{P}(E) \ge 1 - \alpha - o(1)$. □

The proposition is asymptotic with each μ_j fixed in $(0, 1)$, and it is the systems with few passes or few failures that it does not protect. A bootstrap-t interval undercovers for them and collapses to a point when a system passes or fails every task in the sample; a system with true accuracy 0.02 on 50 tasks fails every task a third of the time, and on a small board of four systems the band built from bootstrap-t intervals alone covered the true curve only 62% of the time. For pass/fail outcomes a system with fewer than ten passes or ten failures therefore gets a Clopper–Pearson interval at level $\alpha/(2K)$ in place of the bootstrap interval, and the maximal deviation is taken over the other systems and all costs at level $\alpha/2$ (at level α, with Clopper–Pearson at α/K, when every system is small, and with no Clopper–Pearson part when none is). The Clopper–Pearson half is exact whatever the data select, since each interval covers with probability at least $1 - \alpha/(2K)$ whether or not its system is small, so the K of them fail together with probability at most $\alpha/2$. The maximal-deviation half is asymptotic, and the proposition covers it when every system is regular, with expected passes and failures both tending to infinity, so that no system is small with probability tending to one. For a sparse system, or one whose expected passes lie near ten, the switch between the two intervals is decided by the data, and the total $1 - \alpha$ is a design target supported by the simulations (Table 2), not a theorem. The low-accuracy boards of HAL (AssistantBench, SciCode) are the case this is for. Nothing in the proof uses the cost order, so the band holds where the frontier set of Section 3.3 needs the order to be recoverable. The price is width: a system whose cost is uncertain is allowed into the upper edge long before it is into the lower one. The band is a statement about the curve, the set a statement about its members; the two use different statistics and neither implies the other. The band is what to draw on a leaderboard page, Figure 1; its width on HAL, as a median over the evaluated costs, runs from 6 points (ColBench frontend) to 62 points (CORE-Bench hard), and in simulation it covers the true curve 0.95 to 1.00 of the time in all 15 scenarios, including fixed costs, nearly tied costs and the low-accuracy board (Tables 2 and 3). The coverage is above the nominal level on most boards because the band takes one maximal deviation over all $2K$ means for a curve that depends on one maximum at each cost, so it is conservative by construction. A first version of the band that trusted the sample cost order covered the true curve only 18 to 62% of the time on those boards, which is why the construction above lets a system into an edge only on the strength of its cost interval.

3.3 A confidence set for the frontier

Call j a *rival* of i when $c_j \le c_i$ and $j \ne i$: a rival is any system that could dominate i on cost. For each pair let $\widehat{\text{se}}_{ij}$ be the paired standard error of $\bar{y}_j - \bar{y}_i$, computed from the task-level covariance

of the two outcome series, and let

$$G_{ij} = \frac{\bar{y}_j - \bar{y}_i}{\widehat{\mathrm{se}}_{ij}}$$

be how far j sits above i in standard-error units. Two systems with identical outcomes on every task have a zero paired standard error and a zero gap; we set the standard error to 10^{-12}, so that $G_{ij} = 0$ and the pair never excludes a system. System i is on the observed frontier exactly when $G_{ij} < 0$ for every rival (with the tie conventions). The confidence set keeps i unless some rival beats it by more than a simultaneous margin:

$$\widehat{F}_\alpha = \Big\{ i : \max_{j \text{ rival of } i} G_{ij} \le c_\alpha \Big\}.$$

The margin is a bootstrap quantile. Resample the T tasks with replacement B times; in resample b write $\bar{y}_k^{(b)}$ for the accuracies and, when costs are per task, $\bar{c}_k^{(b)}$ for the costs, so that the rival sets are resampled too. Let

$$M_i^{(b)} = \max_{j \text{ rival of } i \text{ in resample } b} \frac{(\bar{y}_j^{(b)} - \bar{y}_i^{(b)}) - (\bar{y}_j - \bar{y}_i)}{\widehat{\mathrm{se}}_{ij}},$$

the largest centred, studentised deviation of any rival from i, and let $M^{(b)} = \max_i M_i^{(b)}$ be the largest such deviation on the board. The margin is the $(1-\alpha)$ quantile of $M^{(b)}$ over resamples, $c_\alpha = \max(0, q_{1-\alpha}(M))$. This is Hsu's multiple-comparisons-with-the-best construction [13], as implemented for the confidence set of the best in Banahene and Opoku [5], with the rival set of each system cut down to the systems that cost no more and the margin taken jointly over systems. Two properties are immediate: $\widehat{F}_\alpha$ always contains the observed frontier, because a frontier member has every $G_{ij} < 0 \le c_\alpha$; and it shrinks toward the observed frontier as T grows, because the margin is in standard-error units.

Proposition 2 (the set contains the true frontier)**.** *Fix the costs. Let the task vectors be independent and identically distributed with finite variances, with the population variance of $y_{jt} - y_{it}$ positive for every pair compared, and let K be fixed as $T \to \infty$. Let F be the set of systems that no rival beats in true accuracy, the i with $\mu_j \le \mu_i$ for every rival j; it contains the true frontier of Section 2, which also drops a system tied in true accuracy with a strictly cheaper one. Then*

$$\mathbb{P}(F \subseteq \widehat{F}_\alpha) \;\ge\; 1 - \alpha - o(1).$$

Proof. For $i \in F$ and any rival j, $\mu_j - \mu_i \le 0$, so $G_{ij} \le Z_{ij} := \{(\bar{y}_j - \bar{y}_i) - (\mu_j - \mu_i)\}/\widehat{\mathrm{se}}_{ij}$, and therefore $\max_{i \in F} \max_j G_{ij} \le \max_{i \in F} \max_j Z_{ij} \le \max_i \max_j Z_{ij} =: Z$, each maximum over the system's rivals, sets that do not change with the data when costs are fixed. By the multivariate central limit theorem the centred, scaled mean vector is asymptotically Gaussian, and the nonparametric bootstrap of the mean vector is consistent [7]. Z and M both divide by the sample standard errors $\widehat{\mathrm{se}}_{ij}$, which converge in probability to their population values, so by Slutsky's theorem and the continuous mapping theorem Z and the bootstrap M share the limit law of the maximum of the coordinates of a Gaussian vector with positive coordinate variances, whose distribution function is continuous and strictly increasing; hence c_α converges to that law's $(1-\alpha)$ quantile and $\mathbb{P}(Z > c_\alpha) \to \alpha$. Hence $\mathbb{P}(F \not\subseteq \widehat{F}_\alpha) = \mathbb{P}(\max_{i \in F} \max_j G_{ij} > c_\alpha) \le \mathbb{P}(Z > c_\alpha) \le \alpha + o(1)$. □

Remark 1. *This is a confidence set for the frontier as a set: with probability about $1-\alpha$ every truly undominated system is kept at once. Two smaller sets with weaker, per-system guarantees come from the same resamples: the largest per-system quantile, $\max_i q_{1-\alpha}(M_i)$, as the margin, and each*

system's own quantile $q_{1-\alpha}(M_i)$; each keeps a given truly undominated system with probability about $1-\alpha$. Section 5 reports all three against the alternatives a practitioner might reach for instead, and on HAL the three sets hold 280, 259 and 248 systems. The joint set is the one the rest of the paper calls the frontier set.

Proposition 3 (per-task costs)**.** *Let costs be measured per task, with the task vectors of outcomes and costs i.i.d., every accuracy and cost having positive variance and $y_{jt} - y_{it}$ having positive variance for every pair, and let the true mean costs $\gamma_1, \ldots, \gamma_K$ be distinct. Let F be defined from the true accuracies and true mean costs. Then the frontier set built with per-task costs resampled jointly with the outcomes satisfies $\mathbb{P}(F \subseteq \widehat{F}_\alpha) \geq 1 - \alpha - o(1)$. (The band of Proposition 1 needs no such condition.)*

Proof. With distinct true mean costs, the sample mean costs recover the true cost order with probability tending to one, and so do the resampled mean costs within every resample except a vanishing share, since each is $\sqrt{T}$-consistent for its target. On that event every rival set, real and resampled, equals the true rival set, and the statistics coincide with those of the fixed-cost construction with the true rival sets; Proposition 2 then applies. Ties in true mean cost are the case the proof excludes: a rival set that stays random in the limit. Proposition 4 below removes the condition. □

Remark 2. *On HAL the per-task and fixed-cost versions give identical sets on 10 of the 10 boards with per-task costs: cost differences between neighbouring systems are mostly many times their standard error, so on these boards cost uncertainty does not move the set. That is a fact about HAL's cost spread, not a general property, and the next paragraph shows what happens without it. Reruns of one configuration have the same true cost, which breaks the distinctness condition of Proposition 3 for those pairs in the harmless direction: whether the sample order puts a twin among a system's rivals or not, the twin is a true rival, so a wrong order can only withhold evidence from the set.*

When true costs tie. Proposition 3 needs distinct true mean costs, and near ties are what break it. Take a system that is truly cheaper than a rival by one per cent, with the rival truly more accurate. The system is on the true frontier, because the rival does not cost less. But when the rival's sample cost happens to come out lower, the standard set treats the rival as a rival and may exclude the system on its evidence. With true mean costs a few per cent apart and per-task costs with a coefficient of variation of 0.8, the standard set contains the whole true frontier in only 0.26 of simulated boards (Table 3), against the nominal 0.95. The same failure arises when a near-copy of a frontier system costs a few per cent less than the original (0.80, Table 3). The remedy is to exclude a system only on the evidence of rivals that are surely no costlier, and to take the margin over every pair of systems, so that it does not depend on which pairs the cost data happen to order.

Proposition 4 (tie-robust set)**.** *Let costs be measured per task, with the task vectors of outcomes and costs i.i.d., every accuracy and cost having positive variance and $y_{jt} - y_{it}$ having positive variance for every pair, with no condition on ties among the true mean costs. Let c_{cost} be the $(1-\alpha/2)$ bootstrap quantile of the largest of $|\bar{c}_j^{(b)} - \bar{c}_j|/\mathrm{sg}_j$ over systems, and call j a* sure rival *of i when $\bar{c}_j + c_{\mathrm{cost}}\mathrm{sg}_j \leq \bar{c}_i - c_{\mathrm{cost}}\mathrm{sg}_i$. Let c' be the $(1-\alpha/2)$ bootstrap quantile of the largest studentised deviation $\{(\bar{y}_j^{(b)} - \bar{y}_i^{(b)}) - (\bar{y}_j - \bar{y}_i)\}/\widehat{\mathrm{se}}_{ij}$ over every ordered pair $i \neq j$. The tie-robust set keeps i unless some sure rival j of i has $G_{ij} > c'$, and it keeps every system on the observed frontier. With F the set of systems i with $\mu_j \leq \mu_i$ for every $j \neq i$ with $\gamma_j \leq \gamma_i$, the set contains F with probability at least $1 - \alpha - o(1)$.*

Proof. Let E be the event that every γ_j lies in its cost interval $\bar{c}_j \mp c_{\text{cost}}\text{sg}_j$. As in Proposition 1, $\mathbb{P}(E) \geq 1 - \alpha/2 - o(1)$. On E a sure rival j of i has $\gamma_j \leq \bar{c}_j + c_{\text{cost}}\text{sg}_j \leq \bar{c}_i - c_{\text{cost}}\text{sg}_i \leq \gamma_i$, so every sure rival is a true rival. For $i \in F$ and a sure rival j, $\mu_j \leq \mu_i$ and so $G_{ij} \leq Z_{ij}$, with Z_{ij} the centred studentised difference of Proposition 2. Excluding i therefore needs $Z := \max_{i \neq j} Z_{ij} > c'$, a maximum over a fixed index set, and the argument of Proposition 2 gives $\mathbb{P}(Z > c') \leq \alpha/2 + o(1)$. A union bound over E^c and this event gives $1 - \alpha - o(1)$. Adding the observed frontier only enlarges the set. □

The price is size. When costs are tied more closely than their noise, no system is surely cheaper than another, nothing is excluded, and the set is the whole board. That is the honest answer: the data cannot say which systems are cheaper, so they cannot resolve the frontier. The set does not shrink to the true frontier when true costs are exactly tied, since a truly dominated system with an exactly tied dominating twin is kept at any T, the twin being almost never surely cheaper. On HAL the tie-robust set holds 305 systems against 280 for the standard set, and differs from it on 6 of the 10 boards with per-task costs (Table 5). The cost intervals of the tie-robust construction separate 0.65 to 0.96 of the pairs of systems on those boards, against 0.75 in the base simulations and 0.00 in the nearly tied scenario (supplement S7), so HAL's boards are not in the nearly tied regime. The larger set has two causes that we did not separate: the margin taken over every pair at level $\alpha/2$, and pairs that cannot be told apart on cost. On CORE-Bench hard the set is the whole board.

An exact set for any number of tasks. Proposition 2 is asymptotic. A finite-sample companion for fixed costs keeps i unless some rival j beats it in a one-sided exact paired test at level α/m_i, with m_i the number of i's rivals: the sign test on the discordant tasks for pass/fail outcomes, a sign-flip test for rates, which also needs the paired differences to be symmetric under the null. A truly undominated system has no rival that is truly better, so under the task-wise null each rival's test rejects with probability at most α/m_i and the system is kept with probability at least $1 - \alpha$, for any T, by Bonferroni over its rivals; dividing by the number of rival pairs on the whole board instead gives a joint version, which contains the whole true frontier with probability at least $1 - \alpha$ for any T. With per-task costs the rival sets come from the sample cost order, a rival may truly cost more, and the finite-sample guarantee does not extend; the exact sets are then only a comparison, not a guarantee. The price is power: the exact sets are the bootstrap set's conservative cousins, and on HAL they hold 269 (per system) and 291 (joint) systems against 280 (Table 7).

What it would take to resolve a board. A system that is in the set but off the plain frontier is one whose strongest cheaper rival leads it by G standard errors with $G \leq c_\alpha$. The lead in standard-error units grows like $\sqrt{T}$, so with T' tasks it would be $G\sqrt{T'/T}$, and the system leaves the set once $T' > T(c_\alpha/G)^2$. Two summaries of this per board are reported in Section 6: the task count at which half of the currently unexcluded systems would be excluded, and the task count at which a system trailing a cheaper rival by five accuracy points would be excluded, given the board's typical paired noise. Neither is a sample-size formula for a new benchmark, since the gaps on a future board are unknown; they say how far the present boards are from resolving the frontiers they draw.

3.4 Two descriptive summaries

Bootstrap frontier stability. The share of task resamples in which system i is on the observed frontier, with costs resampled when they are per task, describes how stable the system's membership

is under the task process. It is not a probability that the system is truly undominated, and it is not calibrated: with 20 systems on 50 tasks it averages 0.63 over truly undominated systems and 0.10 over truly dominated ones (Table 2). We report it only as a stability measure.

The cheapest unresolved competitor of the best. The confidence set of the best [13] is the set of systems that cannot be excluded as the most accurate, with no reference to cost: the construction of Section 3.3 with every other system as a rival. Its cheapest member, which we call the cheapest unresolved competitor of the best, answers the question a reader asks: which is the cheapest system the data cannot yet rule out as the best? It is not an equivalence claim. A system can be materially worse and still unresolved when the board is small, and non-rejection is not evidence of equality. Its coverage follows from the set's: the true best is in the set with probability at least $1 - \alpha$, and then the cheapest member costs no more than the true best on the observed costs. That the true costs are ordered the same way needs the cost order to be recoverable, which the proof does not supply. In simulation the true best is in the set in 0.99 to 1.00 of boards over all 15 scenarios, and the competitor's true cost is no more than the true best's in 0.98 to 1.00, except when true costs are nearly tied, where it is 0.87 (Table 3).

3.5 Extensions: new entrants and selected winners

A live board changes when submissions arrive, and a system reported as on the frontier was selected for being there. Two results built on the selective inference of the companion paper [5] address those facts. They are in the appendices so that the main text stays about the frontier as an estimated object.

Certifying a frontier entrant (Appendix A). A new system is on the observed frontier exactly when it beats the most accurate system that is cheaper than it, so announcing an entrant is a record claim against one holder. The exact selective test of the companion paper certifies that record claim, with chain-wise error control over the board's history. The record claim is not a claim of Pareto membership, and we prove that no certificate built on the entrant and its strongest rival alone can hold its level against the composite null that some cheaper system dominates the entrant. Certificates of Pareto membership are given under a conditional null, in a normal-approximation form and an exact form for pass/fail outcomes. Walking HAL's boards in submission order, 69 systems were testable entrants, in that they beat every cheaper system on arrival, and another 43 were simply the cheapest so far with no rival to beat. Of the entrants, 6 can be certified as record claims, and 8 as Pareto members in the normal form, under alpha-spending over each board's chain.

The frontier winner's curse (Appendix B). The reported accuracy of a frontier member is inflated by the selection that put it there. The corrections of the companion paper apply with the pool of cheaper systems in place of the board. On HAL they take 1.7 points off a member's reported accuracy on average and up to 9.4.

3.6 What is proved and what is simulated

Table 1 says, for each tool, what is proved under which conditions and what rests on simulation or held-out data.

Tool	Proved	Checked by simulation or held-out data
Frontier band	Covers the whole frontier curve with probability $\geq 1-\alpha$, asymptotically, for fixed or per-task costs and with no condition on ties (Prop. 1); the Clopper–Pearson part for systems with few passes or failures is exact	Coverage of the true curve in all 15 scenarios (Tables 2 and 3)
Frontier set	Contains the whole true frontier with probability $\geq 1-\alpha$, asymptotically, for fixed costs (Prop. 2) and per-task costs without ties (Prop. 3)	Retention against six alternative sets (supplement S11); near-copies, outcome-dependent costs, nearly tied costs and clustered tasks (Table 3); stability on held-out halves (Table 4)
Tie-robust set	Contains the whole true frontier with probability $\geq 1-\alpha$, asymptotically, with no condition on ties (Prop. 4)	Retention and size on nearly tied costs (Table 3); size on HAL (Table 5)
Exact frontier sets	With fixed costs, per-system retention $\geq 1-\alpha$ for any T, and the joint version contains the true frontier with probability $\geq 1-\alpha$ for any T (Section 3.3); no guarantee with per-task costs	Sizes against the bootstrap sets on known truth and on HAL
Bootstrap frontier stability	Nothing; a description	Mean over truly undominated and truly dominated systems
Cheapest unresolved competitor	Follows from the best set's coverage [13], on observed costs	Share of boards on which the true best is in the best set and the competitor costs no more (Table 3)
Entrant certificates (Appendix A)	Record claim: exact selective test under the conditional task-wise null (Prop. 5). Pareto certificate: composite null, normal approximation (Prop. 6); exact for pass/fail outcomes under a conditional null (Prop. 7). No pairwise certificate holds its level (Prop. 8). Chainwise error with costs fixed in advance (Cor. 1)	Level and power with near-copy and independent rivals (Table 8); sensitivity to batching
Winner's curse (Appendix B)	The conditional estimate is exact under normality for a fixed pool [16]	Bias and coverage on known truth (Table 10) and on held-out halves

Table 1: What each tool guarantees. Every guarantee holds under the estimand of Section 2: tasks drawn from the benchmark's task process, inference conditional on the submitted systems.

4 Data: HAL with per-task costs rebuilt

HAL publishes, for every run it hosts, an encrypted trace archive in its Hugging Face dataset (`agent-evals/hal_traces`), about 113 GB across 380 archives at the time of the audit. Each archive holds the run's configuration, its per-task outcomes, its total cost as HAL computed it, and the full call log: every model call the agent made, tagged with the task it served and the tokens it used. The decryption key is the public one in HAL's own harness. We read only the results member of each archive, a few hundred kilobytes to a few hundred megabytes, by HTTP range requests, and reduced each one on the fly to a compact record of outcomes, token counts and costs per task. 5 archives whose results member was too large to decrypt in memory on our machine (the largest 12.8 GB) were skipped; the other 375 were read.

Per-task cost. A task's cost is the priced token usage of the model calls logged under it, with HAL's own formula (fresh input tokens at the model's input price, cache reads and writes at the model's cache price, output tokens at the output price), one price table, HAL's current one, for every run, and each model call counted once. The last clause matters. HAL's reported total sums the usage block of every span in the trace, and an agent that calls a provider through a wrapper library logs two spans for one call, the wrapper and the provider call inside it, both carrying the same usage block. The effect was reported on the dataset's discussion board in September 2026 and confirmed by the HAL team [18]; HAL's pages now carry a caveat, but the totals stand. Of the 332 runs we read with usage, 141 have a reported token total of exactly twice the once-counted total, 44 fall between once and twice, 17 above twice, and 130 are unaffected; over all runs the reported totals overstate token use by 49%. Because the doubling depends on the agent's software stack and not on the model, it can change the cost order of agents on the same board; on the boards where it changes HAL's frontier it moves 4 systems (Table 6). We drop the duplicate spans (a call whose parent carries an identical usage block, or whose children's usage sums to its own), and the supplement lists the ratio for every run. As a check, our per-task costs sum to HAL's reported total to the cent on runs without nested spans that were made after the last price change of the models they used, and to exactly half of it on runs where every provider call was wrapped. Where the two differ otherwise it is because HAL's total embeds the prices in force when the run was made: an o3 run from May 2025 reports several times what the current table gives for the same tokens, o3's price having fallen by four fifths in June 2025, and an early CORE-Bench run priced cached tokens at the full input rate. A frontier compares systems with one another, so it needs one price list and one way of counting; we reprice and recount everything and report HAL's figure beside ours in the released tables. The supplement lists every archive with HAL's total, the ratio of token totals, our recount and the run's status. The two effects compound: over the 318 runs costing at least a dollar, HAL's reported totals sum to $44,679 against $26,782 once counted and repriced, the median run is overstated by a factor of 2.00, 179 runs by more than half, and the largest gap is a factor of 31.

Outcomes and task sets. Outcomes come from the run's lists of successful and failed tasks or from its per-task evaluation results. A benchmark's task set is the tasks with an outcome in at least half of its runs; a run's task without a logged outcome counts as a fail, which touches 9 of the 356 kept runs, each missing at most 5% of its tasks. Reruns of one configuration and HAL's test uploads of the browser agents are separate systems, as HAL lists them; they are each other's rivals at similar cost (a median factor of 1.09 apart, Section 7) and add to the dominated-on-arrival count of Section 6. On ColBench backend programming HAL's accuracy is the share of tasks scored exactly one, and we use that; on ColBench frontend design HAL's metric is a continuous similarity score in $[0, 1]$, and we keep it, so "accuracy" there is a mean score.

				Frontier set			Band	Plain frontier		Stability	
Systems	Tasks	Costs	Corr.	kept, each	kept, all	size (true)	covers	falsely on	missed	true	dominated
8	50	per task	0.6	0.994	0.98	7.1 (4.0)	0.98	12%	17%	0.75	0.15
8	150	per task	0.6	0.991	0.97	6.2 (3.9)	0.98	8%	11%	0.84	0.11
8	500	per task	0.6	0.993	0.97	5.2 (3.9)	0.98	5%	6%	0.91	0.07
20	50	per task	0.6	0.996	0.98	16.9 (5.8)	1.00	19%	28%	0.63	0.10
20	150	per task	0.6	0.992	0.95	13.4 (5.6)	0.98	12%	15%	0.78	0.07
20	500	per task	0.6	0.994	0.96	9.9 (5.7)	0.98	7%	9%	0.86	0.04
20	150	fixed	0.6	1.000	1.00	13.6 (5.9)	0.99	10%	15%	0.79	0.06
20	150	per task	0.3	0.991	0.96	10.5 (4.6)	0.99	11%	15%	0.79	0.05
60	45	per task	0.6	0.999	0.99	49.0 (8.3)	1.00	29%	41%	0.50	0.05
20	65	per task	0.6	0.997	0.98	18.0 (5.9)	1.00	19%	30%	0.61	0.09

Table 2: Known-truth simulation, 300 boards per scenario. Kept, all: share of boards on which every truly undominated system is kept in the frontier set (the guarantee of Proposition 2; target at least 0.95; Monte Carlo standard error at most 0.013). Kept, each: share of truly undominated systems kept, one by one. Size: mean size of the set, with the true frontier's size in brackets. Band covers: share of boards on which the band of Section 3.2 contains the true frontier curve at every evaluated cost. Plain frontier: share of its members that are truly dominated, and share of the true frontier it misses. Stability: bootstrap frontier stability, the share of task resamples in which a system is on the observed frontier, averaged over truly undominated and over truly dominated systems.

Exclusions. Of the 375 run records, 356 enter the audit. 11 runs were dropped because they carried outcomes for fewer than 95% of the benchmark's tasks, and 8 because their call logs carried no model call on more than half of their tasks (for instance runs of Claude Code, which does not log through HAL's tracer), so that their cost is unknown rather than zero; a system with an unknown cost cannot be placed on a cost-accuracy frontier, and dropping it is the only honest option. The supplement lists every dropped and skipped run. A kept run's tasks that have an outcome but no logged model call, a logging gap or a run that failed before its first call, are priced at the run's mean cost over its costed tasks rather than at zero, since a zero would push the run toward the frontier: 47 kept runs have at least one such task, 7 have more than a tenth, and 1.5% of all system-task pairs are imputed; the supplement gives the count per system. On Online-Mind2Web the browser agents' logs carry a call record per step but no token usage on any call, so no per-task cost exists for any run there and there are no per-call usage blocks to double count; the run totals, which HAL computed from the stored run-level usage, reprice exactly under the current table (except for two o3 runs, for the reason above), and we use each run's repriced total spread evenly over its tasks. On that board cost is therefore a fixed number per system, not resampled, and it is marked with a dagger in every table and figure.

Why only HAL. Terminal-Bench 2 publishes cost for three of 244 job files and no per-task results, and SWE-bench Verified's cost metadata covers 47 submissions without per-task results. Neither supports paired inference, so the audit rests on HAL's 11 benchmarks, with per-task outcomes and per-task costs for every kept run on 10 of them.

5 Validation on known truth

Boards were simulated with $K \in \{8, 20\}$ systems and $T \in \{50, 150, 500\}$ tasks, true accuracies and mean costs spread so that several systems are truly undominated, task difficulty shared across

systems so that outcomes are correlated within a task, and per-task costs that vary around each system's mean. On each of 300 boards per scenario the observed frontier and the frontier set were computed from the sample and compared with the true frontier. A share near 0.95 estimated from 300 boards has a Monte Carlo standard error of at most 0.013, so shares from about 0.93 up are consistent with the nominal level. Table 2 has the results for ten scenarios: the six board shapes of the base design and four variants of one of them. Every truly undominated system was kept at once on 0.95 to 1.00 of boards (0.95 to 0.98 for the six base shapes), and a given truly undominated system was kept 0.991 to 1.000 of the time. The plain frontier, over the same boards, wrongly included 5 to 29% of its members and missed 6 to 41% of the true frontier. The price of the guarantee is size: with 20 systems the set held 16.9 systems on average against a true frontier of 5.8 at $T = 50$, 13.4 against 5.6 at $T = 150$, and 9.9 against 5.7 at $T = 500$. The set does not become informative until a board has a few hundred tasks, which is the central practical message of the paper. Four of the ten scenarios probe the assumptions of the base design: costs fixed at their means (the case Proposition 2 covers), a weaker link between cost and ability, a crowded board of 60 systems on 45 tasks, the shape of CORE-Bench hard, and a board of low accuracies, 20 systems on 65 tasks with most of them passing a handful of tasks, the shape of SciCode, where the band needs its exact intervals. With fixed costs the set holds 13.6 systems against 13.4 with per-task costs on the same board shape, and on the crowded board it holds 49.0 of 60 systems against a true frontier of 8.3, with 29% of the plain frontier truly dominated, which is the picture the audit finds on CORE-Bench hard.

Table S11 of the supplement puts the set beside the alternatives on five of the ten scenarios; the ranges below span all ten. The joint margin keeps the whole true frontier 0.95 to 1.00 of the time, and the exact joint set does too, at a larger size. The two per-system variants fall to 0.90 on some boards. The two shortcuts fail where the boards are small or crowded, which is where a leaderboard needs the set most: unadjusted pairwise tests keep the whole true frontier 0.77 to 0.98 of the time, and the frequency rule 0.46 to 0.98, because neither accounts for the number of rivals a system faces. With 20 systems on 50 tasks the sets hold 16.9 (joint), 18.1 (exact joint), 12.3 (unadjusted) and 10.2 (frequency) systems against a true frontier of 5.8.

5.1 Stress tests

The base design leaves out features of real boards that the guarantees lean on, and Table 3 adds them one at a time to the 20-system board of 150 tasks (200 for the clustered one). Near-copy systems: six of the 20 are degraded copies of another system, each turning a pass of its parent into a failure with probability 0.1, as when one agent is run with a smaller model or a lower effort. In one run the copies cost 3% more than their parents, so a copy can only be dominated; in the other they cost 3% less, so a copy is a cheap, highly correlated rival that can itself belong on the frontier. Cost-outcome dependence: failed tasks cost twice as much as passed ones, as when an agent burns tokens on a task it then fails. Nearly tied costs: true mean costs a few per cent apart, closer than the standard error of a mean cost. Clustered tasks: 200 tasks in 40 groups of five that share a difficulty shift and a system-by-group effect, so tasks are not independent draws. Three of the five stresses leave the guarantee where it was. With costlier near-copies the standard set keeps the whole true frontier in 0.98 of boards, with outcome-dependent costs in 0.94 and with clustered tasks in 0.95, against 0.95 on the reference board; resampling the groups instead of the tasks gives 0.96 at a larger size (13.8 systems against 12.7). Clustering at this strength moves the set little, plausibly because group shifts shared by all systems cancel in a paired difference and the system-by-group effect adds only a modest variance; only one clustering strength was tried, so this is a single-point result. The band is more exposed, since it uses each system's own accuracy: it

Scenario	Standard set kept, all	size	Tie-robust set kept, all	size	True frontier	Band covers	Best set true best in	competitor no dearer
Reference: 20 systems, 150 tasks	0.95 (0.012)	13.4	1.00 (0.000)	16.1	5.6	0.98	1.00	0.99
Near-copies	0.98 (0.008)	13.1	1.00 (0.000)	16.4	4.9	0.99	1.00	1.00
Cost-outcome dependence	0.94 (0.014)	12.4	1.00 (0.000)	15.5	5.3	0.98	1.00	0.98
Near-tied costs	0.26 (0.025)	11.1	1.00 (0.000)	20.0	5.7	0.98	1.00	0.87
Clustered tasks	0.95 (0.013)	12.7	1.00 (0.000)	15.4	5.6	0.95	1.00	0.99
groups resampled	0.96 (0.011)	13.8						
Cheaper near-copies	0.80 (0.023)	13.8	1.00 (0.000)	16.9	6.8	0.98	1.00	0.99

Table 3: Stress tests on known truth, 300 boards per row, 20 systems; Monte Carlo standard errors in brackets. Kept, all: share of boards on which every truly undominated system is kept. Standard set: the set of Section 3.3. Tie-robust set: the set of Proposition 4. Best set: share of boards on which the true best system is in the best set, and on which the cheapest unresolved competitor of the best costs no more than the true best. Clustered tasks: the set built by resampling tasks and, in the second row, by resampling the groups.

covers the true curve in 0.95 of the clustered boards.

Two stresses break the standard set, and both are near ties in cost. With nearly tied costs it contains the whole true frontier in 0.26 of boards, and a given truly undominated system in only 0.77. With cheaper near-copies it contains the frontier in 0.80 of boards (a given system in 0.97): a copy that costs 3% less than its parent belongs on the true frontier, but when the sample cost order puts the parent first, the parent becomes the copy's rival, and its paired lead over the copy has a tiny standard error. The tie-robust set keeps the frontier in every board in both scenarios. In the nearly tied one its size is 20.0 of 20: when true costs are closer than their noise no system is surely cheaper than another, and the data cannot resolve the frontier, which the set says by returning the whole board. The band, which does not use the cost order, still covers the true curve in 0.98 of the nearly tied boards, at a median width of 68 points. The cheapest unresolved competitor of the best costs no more than the true best in 0.87 of them, against 0.98 to 1.00 elsewhere, because it also leans on the cost order. The tie-robust set is conservative in the other scenarios as well: it keeps the whole true frontier on every board, at a size 6 to 26% above the standard set's across the other scenarios, and 20 to 25% in Table 3. Whether a real board resembles these cases can be read from the data: the share of system pairs whose cost intervals do not overlap is 0.00 in the nearly tied scenario and 0.75 in the base design, against 0.65 to 0.96 on the HAL boards with per-task costs. Most HAL boards look like the base design in this respect, but on CORE-Bench hard the tie-robust set is the whole board (Table 5), a caution that the cost order there is only partly resolved.

5.2 Held-out halves of the real boards

Simulation has a known truth but a made-up world. Table 4 repeats the frontier check on the real boards with at least 100 tasks, with the tasks split at random into two halves 200 times. The frontier set and the plain frontier are computed on one half, and the systems on the other half's own observed frontier are the target. This is a stability check, not a coverage check. Half B's frontier is itself a noisy estimate, so even a perfect method would not keep all of its members, and what a large set keeps is partly a matter of its size. We therefore read the retention beside a size-matched random baseline. A's frontier set keeps 1.00 of B's frontier members on every board, A's plain frontier 0.60 to 0.99, and the size-matched random baseline 0.77 to 0.99. The set therefore beats random padding of the same size on every board, by 0.01 to 0.23, most where it is much smaller than the board (it holds 50 to 86% of the systems). It is not merely large: it concentrates on the systems that tend to be on a frontier next time. The two halves share the task pool, so the check

Benchmark	Tasks	Systems	Mean set size	B's frontier members kept by A's frontier set	B's frontier members kept by A's plain frontier	B's frontier members kept by size-matched random
ColBench backend	1000	22	16.6	1.00	0.82	0.93
ColBench frontend	100	14	7.0	1.00	0.66	0.77
GAIA	165	36	24.2	1.00	0.60	0.83
Online-Mind2Web†	300	32	27.0	1.00	0.71	0.94
ScienceAgentBench	102	25	21.5	1.00	0.63	0.94
USACO	307	12	7.0	1.00	0.99	0.99

Table 4: Stability check on the HAL boards with at least 100 tasks, 200 random half splits each. The frontier set and the plain frontier are computed on half A; the systems on half B's own observed frontier are the target. Columns: the share of those systems kept by A's frontier set, by A's plain frontier, and by a size-matched random baseline (A's plain frontier plus randomly chosen systems up to the size of A's frontier set); mean size of A's set.

Benchmark	Systems	Tasks	Observed frontier	Frontier set (share)	Tie-robust set	Best set	Top system (%, $)	Cheapest unresolved (%, $)
assistantbench	32	33	3	32 (100%)	32	32	15.2, 4.39	3.0, 0.02
colbench backend programming	22	1000	9	14 (64%)	15	5	43.7, 0.20	38.6, 0.02
colbench frontend design	14	100	4	7 (50%)	7	5	82.9, 0.12	81.6, 0.03
corebench hard	60	45	8	53 (88%)	60	1	75.6, 6.08	75.6, 6.08
gaia	36	165	10	22 (61%)	26	7	74.5, 1.13	64.2, 0.74
online mind2web†	32	300	6	23 (72%)	23	16	42.3, 0.57	36.3, 0.30
scicode	37	65	3	37 (100%)	37	37	9.2, 0.72	1.5, 0.00
scienceagentbench	25	102	4	21 (84%)	21	17	33.3, 0.06	27.5, 0.02
swebench verified mini	39	50	6	26 (67%)	33	11	72.0, 1.57	54.0, 0.67
taubench airline	47	50	7	40 (85%)	45	26	66.0, 3.14	44.0, 0.04
usaco	12	307	4	5 (42%)	6	1	69.7, 0.11	69.7, 0.11

Table 5: Frontier audit of HAL's benchmarks at $\alpha = 0.05$, $B = 4000$ task resamples. Observed frontier: systems no cheaper system beats on the point estimates. Frontier set: systems that cannot be excluded from the frontier; share: that count over the systems on the board. Tie-robust set: the set of Proposition 4. Best set: systems that cannot be excluded as the most accurate. Top system: the most accurate system; cheapest unresolved: the cheapest unresolved competitor of the best, each with its accuracy and its cost per task in dollars. Costs are dollars per task, repriced from the call logs with one price table. †: no per-task costs exist on this board; each run's repriced total is spread evenly over its tasks and treated as fixed, so there is no cost uncertainty and the tie-robust set is defined to be the frontier set.

covers task-sampling noise only, which is the noise the method models. The winner's-curse part of the same split is in Appendix B.

6 Audit of the Holistic Agent Leaderboard

HAL's own frontiers change under consistent costing. Before any uncertainty enters, the point-estimate frontier itself depends on how costs are counted and priced. Here "HAL's published frontier" means the frontier of the kept runs under HAL's reported totals; the runs we dropped, such as those of Claude Code, are absent from it and may sit on HAL's own. Table 6 draws it three ways. Going from HAL's reported totals to once-counted, repriced costs changes the frontier on 8 of 11 boards: 10 systems leave it and 18 join, 28 changes against 56 systems on the reported frontiers (50%). Removing the duplicate spans alone moves 4 systems on 2 boards, because the doubling is shared by the agents that use the same software stack and mostly leaves their order intact. Pricing every run under one table moves 24 more, because HAL's totals carry the prices of the month each run was made, and because 5 runs are reported at a total near zero although their call logs price at more than a cent per task, all of them sitting on HAL's frontier for that reason alone. A reader of HAL's pages is therefore looking at frontiers that mix two counting rules and a

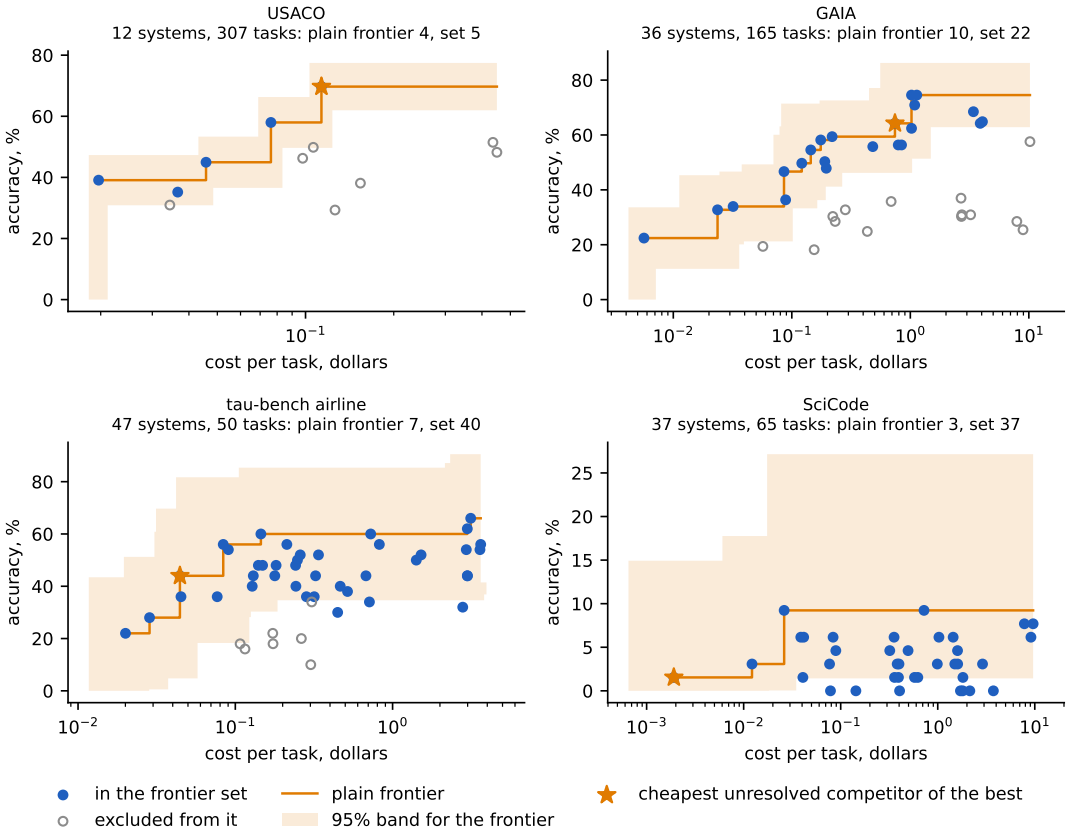


Figure 1: Cost per task against accuracy on four of the eleven HAL benchmarks: USACO, where the set is close to the plain frontier, GAIA and tau-bench airline, where it is much wider, and SciCode, where nothing can be excluded. The orange staircase is the observed frontier. Filled points are systems that cannot be excluded from the frontier at the 5% level; hollow points can be. The star is the cheapest unresolved competitor of the best: the cheapest system that cannot be excluded as the most accurate. The shaded region is the 95% simultaneous band for the frontier curve (Section 3.2). Costs of zero are drawn at 10^{-4} dollars. All eleven boards are in the supplement.

Benchmark	Plain frontier size under reported	once-counted	ours	Reported to ours lost	gained	Changes due to counting	pricing
assistantbench	3	3	3	1	1	0	2
colbench backend programming	9	9	9	0	0	0	0
colbench frontend design	3	3	4	0	1	0	1
corebench hard	5	5	8	2	5	0	7
gaia	5	5	10	1	6	0	7
online mind2web†	6	6	6	0	0	0	0
scicode	3	3	3	1	1	0	2
scienceagentbench	8	6	4	4	0	2	2
swebench verified mini	4	6	6	0	2	2	0
taubench airline	6	6	7	1	2	0	3
usaco	4	4	4	0	0	0	0
All	56	56	64	10	18	4	24

Table 6: The plain frontier of each board under three costings of the same kept runs: HAL's reported totals, those totals with the duplicate spans removed (HAL's historical prices kept), and the once-counted, repriced costs of the audit. Lost and gained: systems that leave or join the frontier going from HAL's costing to ours; the last two columns split the changes into those made by counting each call once and those made by repricing after that.

year of price changes. The supplement draws the two boards that change most.

Table 5 and Figure 1 give the audit proper; the set sizes are also drawn in the supplement. Six readings.

The frontier is far less certain than the picture suggests. Across the 11 benchmarks 64 of 356 systems sit on the observed frontier, and 280 cannot be excluded from it. A pooled share hides the spread, so Table 5 gives the share board by board. It runs from 42% to 100%, and without the 2 boards on which nothing can be excluded the pooled share is 74%. The simultaneous margin is 3.04 to 3.77 standard errors of a paired difference, and on a board of 50 tasks a paired standard error is several accuracy points, so a system is ruled off the frontier only when a cheaper one leads it by ten to fifteen points.

The band says the same thing as a curve. The shaded region in Figure 1 is the 95% band of Section 3.2. Its half-width at a frontier member is 2.90 to 3.67 of that member's own standard error, and its width, taken as the median over the costs at which it is evaluated, runs from 6 points on ColBench frontend to 62 points on CORE-Bench hard. A reader who wants one picture of what a board can and cannot say about the best accuracy available at each price has it here, and the picture is humbler than the staircase. Even the best-measured boards, ColBench backend with 1000 tasks and USACO and Online-Mind2Web with about 300, have bands of 10 to 18 points; ColBench frontend is narrow (6 points) only because its continuous scores vary little from task to task; and on the boards with 33 to 65 tasks the band is 26 to 62 points wide, a large part of the whole range of accuracies. A step of the staircase can be trusted only where it is taller than the band.

Task count decides what can be said. The two boards where the set is closest to the observed frontier are USACO (307 tasks: frontier 4, set 5) and ColBench backend programming (1000 tasks: frontier 9, set 14). The two boards where nothing can be excluded are AssistantBench (33 tasks, accuracies up to 15.2%) and SciCode (65 tasks, up to 9.2%): with success rates that low and

Benchmark	Tasks	Systems kept						Tasks needed	
		Plain frontier	joint set	largest-margin	per-system	exact, per system	exact, joint	to exclude half	for a 5-point lead
assistantbench	33	3	32	32	32	32	32	171	442
colbench backend programming	1000	9	14	12	12	12	15	2,857	1,012
colbench frontend design	100	4	7	7	7	7	7	7,863	52
corebench hard	45	8	53	49	49	52	57	408	1,630
gaia	165	10	22	20	20	21	23	606	1,699
online mind2web†	300	6	23	21	19	20	24	1,811	1,370
scicode	65	3	37	37	37	37	37	265	365
scienceagentbench	102	4	21	18	18	20	21	529	1,169
swebench verified mini	50	6	26	23	17	24	27	122	1,179
taubench airline	50	7	40	35	32	39	42	291	1,748
usaco	307	4	5	5	5	5	6	1,057	697

Table 7: Five frontier sets and two resolution numbers per board. Joint set: the confidence set of Section 3.3; largest-margin and per-system: its two per-system variants; exact, per system and exact, joint: the finite-sample sets of Section 3.3. Tasks to exclude half: the task count at which half of the systems now in the set but off the plain frontier would leave it, if their gaps to their strongest cheaper rival stayed as observed. Tasks for a 5-point lead: the task count at which a system trailing a cheaper rival by five accuracy points would be excluded, given the board's median paired noise.

samples that small, no cheaper system can be shown to beat any other. CORE-Bench hard, with 60 systems on 45 tasks, keeps 53 of them in the set.

How far the boards are from resolving their frontiers. Table 7 turns the set sizes into task counts. Excluding half of the systems that now sit in the set but off the plain frontier would take 122 to 7,863 tasks across the boards that have such systems; excluding a system that trails a cheaper rival by five points would take 52 to 1,748. Like for like, the exact sets are the larger ones: the exact joint set holds 291 systems against the confidence set's 280 (larger on 7 boards, smaller on 0), and the exact per-system set 269 against the per-system set's 248, which is the usual trade between a finite-sample guarantee and power. The two per-system variants of the confidence set hold 259 and 248 systems against its 280: the joint guarantee costs a few dozen systems across the audit. The two boards with the largest "tasks to exclude half" (ColBench backend and frontend) are the ones where the few unexcluded systems are near-ties with their rivals, which no task count resolves quickly.

The best is often clear even when the frontier is not. The best set is a single system on 2 boards (CORE-Bench hard and USACO) while the frontier set on those boards is larger. A system's rivals for the best set are all the other systems, and for the frontier set only the cheaper ones, so the frontier set is the larger on every HAL board, and a cheap, weak system stays in it whenever no cheaper system beats it by the margin. The nesting is an observation about these boards and not a guarantee, since the two margins are different quantiles. The frontier set is a statement about every system's own neighbourhood of cheaper rivals, and on most boards those neighbourhoods are thinly measured.

The cheapest unresolved competitor of the best is usually not the top system. On 7 of 11 boards the cheapest system that cannot be ruled out as the best is a different and cheaper system than the most accurate one; on 2 further boards (AssistantBench and SciCode) the best set is the whole board, so the cheapest unresolved competitor is simply the cheapest system and the statement is empty. The gaps are large: on tau-bench airline the top system costs $3.14 per task and scores 66.0%, while the cheapest unresolved competitor costs $0.04 and scores 44.0%; on GAIA the pair is $1.13 at 74.5% against $0.74 at 64.2%; on SWE-bench Verified mini, $1.57 at

72.0% against $0.67 at 54.0%. This does not say the cheap system is as good. It says that 50 tasks cannot show that it is worse, at a level that accounts for all 47 systems being compared at once, and more tasks are what would settle it.

How much the remaining costing rules matter. Two further choices in the costing could have driven the results: pricing cache reads and writes at the table's cache price, and imputing a task with no logged model call at the run's mean cost. The supplement recomputes the boards from the stored token counts under the opposite choice for each, the full input price for cached tokens and a cost of zero for an unlogged task. Pricing cached tokens at the full input price raises the mean cost per task by up to a factor of 2.0, on SWE-bench Verified mini, and moves 8 systems on or off the plain frontier on 3 boards and 7 in or out of the frontier set. Costing an unlogged task at zero moves 3 systems on the plain frontier and 2 in the set. The cache-price rule therefore moves more systems on the plain frontier than the double count does by itself (4), though far fewer than the 24 moved by pricing at the prices of each run's own date. The set sizes, which are the audit's main result, change by a few systems.

7 Limitations

Run-to-run noise. Each system has one run. The standard error of a single run is computed from the sample variance of its task outcomes, so it already contains the independent run-to-run noise of each task. What is missing is any excess beyond that: a run that is better or worse than its configuration's average for reasons shared across its tasks. HAL holds 8 pairs of runs of one configuration (same agent, model and settings) on 3 boards, which measure the excess directly. The two runs of a pair differ by 6.4 accuracy points on average and 14.0 at most, and in cost by a median factor of 1.09 (at most 1.72). If every task's outcome flipped independently between the two runs at the observed rate of disagreement, the root-mean-square difference would be 5.6 points; the observed value is 8.3, an excess standard deviation of 4.4 points per run. With 8 pairs this is weak evidence (chi-square 13.5 on 8 degrees of freedom, $p = 0.096$), and it is not nothing. Carried to every board through the per-task disagreement rate and added to both systems of every paired comparison, an excess of that size would grow the frontier sets from 280 to 324 systems in all. If the excess is real, the audit's sets are lower bounds. The remedy is repeated runs, which the method accommodates by averaging them.

Task clusters. Tasks are treated as exchangeable draws. Where tasks cluster (several from one repository, one website, one problem family) the standard errors are optimistic, as in any task-level analysis. In simulation with 40 groups of five tasks that share a difficulty shift and a system-by-group effect, the set built by resampling tasks contains the whole true frontier in 0.95 of boards and the set built by resampling groups in 0.96 (Table 3). The module can resample task groups, but HAL's boards offer little to resample: SWE-bench Verified mini's 50 tasks come from 2 repositories and USACO's 307 from 4 difficulty divisions, so margins of 3.36 and 5.52 standard errors against 3.66 and 3.04 say more about having two or four units than about dependence. Online-Mind2Web (tasks within websites) and ScienceAgentBench (tasks within source papers) probably have many small groups, but we did not group their tasks and so ran no cluster bootstrap on them.

The rest. Costs depend on prices, which change; we fix one table and say which, and a reader pricing a run under another table should rebuild the frontier rather than reuse ours. The recount rests on HAL's published call logs, and we reran no agent. The certificates and the winner's-curse

corrections fix each system's cost at its mean, so selection through noisy costs is not corrected for. Systems may have been tuned on the public tasks, which inflates every accuracy on the board and is outside the model. Outcomes are taken as HAL recorded them; HAL has since disclosed that on tau-bench airline its Generalist agent's episodes that never terminated were recorded as failures, so those rows are lower bounds on that scaffold's accuracy [18]. The guarantees of the band and the set are asymptotic. The band needs no condition on the costs. The set is proved for fixed costs and for per-task costs whose true means are distinct, and the tie-robust set removes that condition at the price of size. On Online-Mind2Web the cost is a run total treated as fixed. The set uses one margin for the whole board, which is conservative when the systems are many; per-system margins would give a smaller set with a weaker guarantee. Finally, the set is a statement about K specific systems on T specific tasks, and a new system or a new task changes it.

8 Recommendations

1. Report the frontier set, or the band, beside the frontier. A board that draws a frontier from point estimates should say which systems cannot be excluded from it, at a stated level, how many tasks that statement rests on, and how many it would take to resolve it (Table 7: from about a hundred to several thousand tasks on the present boards).

2. Publish per-task costs and per-call token counts, count each model call once, and price every run under one named table with its date. The confidence set needs per-task costs to carry the cost uncertainty, and the audit shows that run totals alone can hide a several-fold difference in price between two dates.

3. Where agents built on one model have nearly tied costs, use the tie-robust set, which stays valid when costs tie and says so by growing.

4. Report the cheapest unresolved competitor of the best, with its level, beside the top system. Read it as a reason to run more tasks, not as a purchasing recommendation.

5. Certify frontier entrants with an error budget spent over the calendar, and say which claim is certified: the record claim against the most accurate cheaper system (exact), or Pareto membership (the composite test, under its conditional null), with costs fixed on arrival (Appendix A).

9 Conclusion

A cost-accuracy frontier is an estimate. Treated as one, with a confidence set in place of a line, HAL's benchmarks say less than the lines suggest and more than nothing: on some boards the frontier is sharp, on most it is wide, and on a few it carries no content until more tasks are run. The tools here let a maintainer say which. They hold for tasks drawn from the benchmark's task process, conditional on the systems submitted, and they say nothing about tuning on public tasks or about new kinds of tasks.

Reproducibility

Code, the per-system result summaries, the simulations and the audit are in the paper's public repository (`https://github.com/jopoku16/frontier-leaderboards`), and the methods are in

the open-source package `honestboard` (module `frontier`, version 0.5.1). The compact per-task records derived from HAL's trace archives and the raw archives are not redistributed, because the dataset states no licence; the scripts rebuild them from the public archives. The supplement lists every archive with its status.

A Certifying a frontier entrant

This appendix gives the certificates for a new frontier entrant, the proofs that go with them, and their behaviour on HAL's own timeline.

A.1 Ingredients from the companion paper

Five pieces of Banahene and Opoku [5] are used below, each stated here in the form needed.

The selective exact p-value. For a holder a and a challenger b on the same T tasks with pass/fail outcomes, let n_b be the number of tasks b passes and a fails, n_a the reverse, and $n = n_a + n_b$. A leaderboard tests b only because $\bar{y}_b > \bar{y}_a$, that is $n_b > n_a$. The selective exact p-value is the tail of the sign test conditional on that event, $p_{\mathrm{sel}} = \mathbb{P}(X \geq n_b \mid X > n/2)$ with $X \sim \mathrm{Bin}(n, 1/2)$. For outcomes that are rates in $[0, 1]$ the same conditioning is applied to a sign-flip test of the paired differences by Monte Carlo over 200,000 sign patterns.

The conditional task-wise dominance null. The null under which p_{sel} is valid: on every task, given the two systems' outcomes on the other tasks, b is no more likely to pass than a, with tasks independent. It is the natural "b is not better than a" for paired data. Proposition 1 of the companion paper shows that under this null p_{sel} is super-uniform given the record event, provided the holder was chosen by an event that is increasing in the holder's own outcomes and does not involve b.

The per-date factor and the best-first order. Submissions that share a date are ordered best first, so the challenger tested on a date is the best of that date, a second selection the record event does not cover. The p-value is multiplied by g, the number of submissions on the challenger's date, a Bonferroni factor over the date's members, which restores the guarantee.

Alpha-spending. The t-th tested step is tested at level $\alpha_t = 6\alpha/(\pi^2 t^2)$. The levels sum to α, so by the union bound the probability of certifying any null step over the whole chain is at most α, for any number of steps and any dependence, provided each step's p-value is valid given that the step occurred and the step's level was settled before its outcomes were looked at.

The online fallback. The t-th step is tested at α_t plus the level of the previous step when that step was certified [27, 28], so a run of certified steps accumulates budget. The family-wise guarantee is the same, by a disjointness argument over the runs of certified steps that feed each null step.

A.2 The certificates

A live board changes when submissions arrive. A new system b lands on the frontier when it beats every system cheaper than it; the maintainer who then tests it has selected it for exactly that, and a test that ignores the selection is biased. The frontier event looks harder than the record event because it is an intersection of events against several systems. As an event, it is not.

Let $C(b) = \{j \text{ earlier} : c_j \leq c_b\}$ be the cheaper-or-equal systems when b arrives, and let j^* be the most accurate member of $C(b)$, with ties broken by a rule that does not look at b. Then

$$b \text{ is on the observed frontier on arrival} \quad \Longleftrightarrow \quad \bar{y}_b > \bar{y}_{j^*},$$

because beating the most accurate cheaper system is the same as beating every cheaper system (a tie with a strictly cheaper system leaves b dominated, a tie at equal cost leaves both on the frontier, and b counts as an entrant only when the inequality is strict). The frontier event is therefore the record event of the companion paper for the single pair (holder j^*, challenger b), and the holder was chosen by an event, "j^* is the most accurate system in $C(b)$", that is increasing in j^*'s own outcomes task by task, depends otherwise only on the other cheaper systems, and does not involve b. These are the conditions of that paper's Proposition 1.

Proposition 5 (frontier entrant)**.** *Fix the costs in advance of the outcomes. Under the conditional task-wise dominance null for the pair (j^*, b) and independence across tasks, the selective exact p-value $p_{\text{sel}}(b \text{ vs } j^*)$ is super-uniform conditional on the event that b is on the observed frontier on arrival. A rejection at level α certifies that b beats the system that was the most accurate cheaper system when b arrived.*

Remark 3 (what the certificate says, and what it does not)**.** *The null tested is pairwise. If some cheaper system k truly dominates b while the observed best cheaper system j^* is a weak relative of b (the same agent with a smaller model, a lower effort setting, or a rerun, whose outcomes are correlated with b's), then b can beat j^* with the stated guarantee and still be truly dominated by k. The test is not of level α for the composite null "some cheaper system dominates b". In simulation with such a near-copy as the apparent rival and a truly dominating independent system in the pool, a dominated entrant is certified in 3.3 to 34.8% of the cases in which it lands on the frontier, against 1.1 to 3.6% when every cheaper system is independent of it (20000 boards per case). The certificate is therefore a record claim restricted to the cheaper pool, and it becomes a certificate of Pareto membership only under an added assumption: that whenever b is truly dominated by a cheaper system, it is task-wise dominated by the observed best cheaper system as well. Testing b against each cheaper system with the pairwise selective test and requiring every test to reject does not repair this, because each pairwise test conditions on its own event and not on the whole frontier event. Conditioning every test on the whole event does, at the price of a normal approximation; that is the next proposition.*

Proposition 6 (Pareto certificate)**.** *Let the vector of accuracies be approximately normal, $\bar{y} \sim N(\theta, \Sigma)$ with Σ the task-level covariance divided by T, and let $E = \{\bar{y}_b \geq \bar{y}_k \text{ for every } k \in C(b)\}$ be the frontier event. For each $k \in C(b)$ let p_k be the polyhedral selective p-value of Lee et al. [16] for $H_k : \theta_b \leq \theta_k$ conditional on E, with the lattice correction of the companion paper. Then $p_{\text{IU}} = \max_{k \in C(b)} p_k$ is super-uniform, conditional on E, under the composite null "some system in $C(b)$ truly dominates b", and rejecting when $p_{\text{IU}} \leq \alpha$ certifies at level α that no cheaper-or-equal system is truly at least as accurate as b: that b is on the true frontier of its pool.*

Proof. Under the composite null some H_k holds. E is a polyhedron in $\bar{y}$, so by the polyhedral lemma p_k is uniform conditional on E when $\theta_b = \theta_k$ and super-uniform when $\theta_b < \theta_k$. Hence $\mathbb{P}(p_{\text{IU}} \leq \alpha \mid E) \leq \mathbb{P}(p_k \leq \alpha \mid E) \leq \alpha$. An intersection-union test needs no correction for the number of rivals [6]. □

Two things sit outside the proof. The covariance is a plug-in estimate, and the lattice correction is a continuity adjustment for accuracies that live on the grid $1/T$: it relaxes each constraint by half a step and takes the tail from half a step below the observed value, one part conservative and one not. The null rows of Table 8, at most 4.2% and 4.4%, are their check. The two certificates answer different questions and have different prices. The pairwise test is exact for any T and asks whether b beat the system it displaced; the composite test is exact only under normality with a plug-in covariance and asks whether b beats every cheaper system at once, conditioning on an event that truncates the statistic more. Table 8 shows both on the three simulated worlds: with a near-copy rival the pairwise test certifies a truly dominated entrant in 3.3 to 34.8% of the cases in which it lands on the frontier, the composite test in 1.6 to 4.2%; with independent rivals the pairwise test certifies 1.1 to 3.6% and the composite test 1.7 to 4.4%; and when the entrant is truly better than every cheaper system by half a logit the pairwise test certifies it 10 to 76% of the time and the composite test 15 to 78%. The composite test gives up nothing in power for its stronger claim: the exact sign test is conservative because its statistic is discrete, which shows in its null rates (1.1 to 3.6% against the composite test's 1.7 to 4.4%), and the normal statistic has no such step. What the composite test gives up is exactness. The audit reports both.

Proposition 7 (exact Pareto certificate)**.** *Fix the costs; outcomes are pass/fail. For $k \in C(b)$ let $\mathcal{G}_k$ be the information in the unordered pairs $\{y_{bt}, y_{kt}\}$ and in the outcomes of all other systems, and let H_k be the*

World	Rival shape	Tasks	Boards on frontier	Certified at 5% pairwise	composite	exact composite
independent	independent rivals	50	7802	1.1%	2.1%	1.5%
independent	independent rivals	100	8128	1.1%	1.7%	1.1%
independent	independent rivals	150	9259	2.5%	3.3%	2.8%
independent	independent rivals	300	9619	3.6%	4.4%	3.8%
near-copy	near-copy, flip 0.2	50	7757	3.4%	1.6%	0.9%
near-copy	near-copy, flip 0.2	100	8035	5.1%	1.9%	1.3%
near-copy	near-copy, flip 0.2	150	9131	3.3%	3.6%	2.9%
near-copy	near-copy, flip 0.05	300	9661	34.8%	4.2%	1.8%
better	entrant truly better	50	16377	9.8%	14.7%	11.6%
better	entrant truly better	100	18601	23.7%	29.6%	25.0%
better	entrant truly better	150	19319	41.3%	46.5%	42.8%
better	entrant truly better	300	19931	75.9%	78.3%	76.7%

Table 8: What the two certificates control, 20000 boards per row. In the two null worlds a cheaper independent system with the entrant's true accuracy dominates it; the near-copy rows add a degraded copy of the entrant to the pool, with flip rate 0.2 at 50 to 150 tasks and 0.05 at 300 tasks, the closer copy being the case the impossibility result is about. In the last block the entrant is truly better than every cheaper system by half a logit. Rates are over the boards on which the entrant landed on the observed frontier. A valid Pareto certificate must stay at or below 5% in the null rows.

statement that tasks are independent and, on every task with $y_{bt} \neq y_{kt}$, $\mathbb{P}(y_{bt} = 1 \mid \mathcal{G}_k) \leq \frac{1}{2}$; equality is the sharp null that b and k are exchangeable given the others. Let D_k be the number of tasks on which b and k differ, N_k the number b wins, n_{11} the number both pass, and

$$n_{0k} = \max\Big\{\lceil D_k/2 \rceil,\ \lceil T \max_{j \in C(b)\setminus k} \bar{y}_j - n_{11} \rceil\Big\}, \qquad p_k = \frac{\mathbb{P}(X \geq N_k)}{\mathbb{P}(X \geq n_{0k})}, \quad X \sim \mathrm{Bin}(D_k, \tfrac{1}{2}).$$

On the frontier event $E = \{\bar{y}_b \geq \bar{y}_j \text{ for every } j \in C(b)\}$: (i) under H_k, $\mathbb{P}(p_k \leq u \mid \mathcal{G}_k, E) \leq u$ for every u, with equality at the attainable values under the sharp null; (ii) if H_k holds for some $k \in C(b)$, then $\mathbb{P}(\max_k p_k \leq \alpha \mid E) \leq \alpha$, for any number of tasks. For outcomes in $[0,1]$ that are not pass/fail the same construction by random swaps is valid under the sharp null only.

Proof. Given $\mathcal{G}_k$ the unordered pair on each discordant task is fixed, so N_k is a sum of D_k independent indicators with success probabilities $q_t \leq \frac{1}{2}$ under H_k, and $\mathrm{Bin}(D_k, \frac{1}{2})$ under the sharp null. Swapping b's and k's outcomes on a set of tasks leaves $\bar{y}_b + \bar{y}_k$ and every $\bar{y}_j$ with $j \notin \{b, k\}$ unchanged, so given $\mathcal{G}_k$ the event E is the event $\{N_k \geq n_{0k}\}$ with n_{0k} a function of $\mathcal{G}_k$: the first term of the maximum comes from $\bar{y}_b \geq \bar{y}_k$, the second from $\bar{y}_b \geq \bar{y}_j$ for the other systems and is omitted when $C(b) = \{k\}$. Under the sharp null the law of N_k given $\mathcal{G}_k$ and E is $\mathrm{Bin}(D_k, \frac{1}{2})$ truncated to $\{N \geq n_{0k}\}$, and p_k is the tail probability of that law at the observed value, hence exactly uniform on its attainable values. Under H_k the probability mass function of N_k relative to that of $\mathrm{Bin}(D_k, \frac{1}{2})$ is proportional to $e_n(r)/\binom{D_k}{n}$, with e_n the elementary symmetric polynomial and $r_t = q_t/(1-q_t) \leq 1$; with $S_n = e_n(r)/\binom{D_k}{n}$, Maclaurin's inequality gives $S_{n+1}^{1/(n+1)} \leq S_n^{1/n}$, hence $S_{n+1} \leq S_n \cdot S_n^{1/n} \leq S_n$ because $S_n \leq 1$ for $r_t \leq 1$; the likelihood ratio is therefore nonincreasing, so N_k is smaller than $\mathrm{Bin}(D_k, \frac{1}{2})$ in the likelihood-ratio order, an order that truncation to $\{N \geq n_{0k}\}$ preserves; hence $\mathbb{P}(p_k \leq u \mid \mathcal{G}_k, E) \leq u$, which is (i). For (ii), E is the same event for every k; if H_{k_0} holds then $\{\max_k p_k \leq \alpha\} \subseteq \{p_{k_0} \leq \alpha\}$, and averaging (i) over $\mathcal{G}_{k_0}$ given E gives the bound. An intersection-union test needs no correction for the number of rivals [6]. □

The p-value is the sign test of the companion paper with a higher truncation point, $n_{0k} \geq \lceil D_k/2 \rceil$, and needs no normality, covariance estimate, lattice correction or Monte Carlo: only pass/fail outcomes, and that the null be read as conditional on the other systems' outcomes. That last point is the price. A system

that truly dominates b in means need not satisfy H_k: in the near-copy world b wins every discordant task against the independent dominating system on which the copy passes, so $\mathbb{P}(y_{bt} = 1 \mid \mathcal{G}_k) = 1$ there, and the proof does not cover that world. Table 8 is the evidence for it. In the audit's three simulated worlds the exact certificate certifies a dominated entrant 0.9 to 2.9% of the time with a near-copy rival and 1.1 to 3.8% with independent ones, and a truly better entrant 12 to 77% of the time, on 20000 boards per row. The guarantee is conditional on the weak event E ($\geq$); certification also requires $\bar{y}_b > \bar{y}_j$ for every j, which can only reduce the probability of certification.

Proposition 8 (no pairwise certificate holds its level)**.** *Let ϕ_T be a test of the composite null that b is dominated by some cheaper system, depending on the data only through the outcomes of the entrant b and of the observed most accurate cheaper system j^*, and suppose there is $\varepsilon(m) \to 0$ such that ϕ_T rejects with probability at least $1 - \varepsilon(m)$ whenever b wins every task on which b and j^* differ and there are at least m such tasks. Then there are configurations with $\mu_b \in (0, 1)$ in which b is truly dominated by a strictly cheaper system, the frontier event has probability tending to $\frac{1}{2}$, and the conditional rejection probability given the frontier event tends to one.*

Proof. Put in $C(b)$ a strictly cheaper system k with $\mu_k = \mu_b$, independent of b, and a cheaper copy j of b whose passes are flipped to failures independently with probability $\delta_T = T^{-3/4}$. Then k dominates b. Let $F \approx \mu_b \delta_T T = \mu_b T^{1/4} \to \infty$ be the number of flipped passes. Then $\bar{y}_b > \bar{y}_j$ with probability tending to one, and $\bar{y}_b > \bar{y}_k$ with probability tending to $\frac{1}{2}$, so $\mathbb{P}(E) \to \frac{1}{2}$. Given E, $j^* = j$ exactly when $\bar{y}_b - \bar{y}_k \geq F/T$; the left side is of order $T^{-1/2}$ with a continuous positive limit density at 0^+ while F/T is of order $T^{-3/4}$, so $\mathbb{P}(j^* = j \mid E) \to 1$. Against j, b wins all F discordant tasks and loses none, and $F \to \infty$ in probability. For each fixed m, $\liminf_T \mathbb{P}(\phi_T \text{ rejects} \mid E) \geq 1 - \varepsilon(m)$, because given E both $F \geq m$ and $j^* = j$ hold with probability tending to one; letting $m \to \infty$ gives the claim. □

The class is not vacuous: any test that rejects once b has won a fixed number of discordant tasks and lost none belongs to it, and the selective sign test is one. What lies outside it is a test that needs on the order of $\sqrt{T}$ wins before it rejects, the number an independent rival's sampling noise produces, and such a test pays for its protection by failing to certify a genuine entrant who beats j^* by less than that. The convergence is so slow, of order $T^{-1/4}$, that the proposition describes the shape of the problem and not its size; Table 8 gives the size at 50 to 300 tasks. A leaderboard that lists an agent twice, with a stronger and a weaker model or with two reasoning-effort settings, holds a pair of this kind in a weaker form: HAL's own reruns of one configuration agree on 72 to 92% of tasks, far from the asymptotic construction, which needs agreement on all but about $T^{1/4}$ tasks, and the simulations show what pairs that close do at realistic sizes. Any certificate of Pareto membership must therefore look at the whole cheaper pool, which is what Propositions 6 and 7 do.

Corollary 1 (a chain of entrants)**.** *Let the step index t count the dates on which at least one entrant arrived, and test every entrant of the t-th such date at level α_t (alpha-spending) or at α_t plus the previous such date's level when every entrant of that date was certified (fallback), with its p-value multiplied by its per-date factor g. Then the family-wise error of the chain of certificates is at most α, in the sense of Proposition 5 when the pairwise p-values are used and in the sense of Proposition 6 when the composite ones are.*

Proof. A date's level is a function of which earlier dates had entrants and, for the fallback, of their outcomes, and of no outcome of the date itself; it is predictable, and the null of each proposition is read conditionally on all earlier systems' outcomes. For a null submission s on date d, $\mathbb{P}(s \text{ certified}) = \mathbb{P}(E_s)\mathbb{P}(p_s \leq \alpha_t/g \mid E_s) \leq \mathbb{P}(E_s)\,\alpha_t/g$, by Proposition 5 or 6; summing over the at most g submissions of the date, the expected number of certified null entrants on the date, given the past, is at most $\alpha_t \mathbb{P}(\text{an entrant arrives} \mid \text{past}) \leq \alpha_t$. The step index t_d is predictable and takes distinct values on distinct dates with entrants, so the expected number of certified null entrants over the whole chain is at most $\sum_t \alpha_t = \alpha$. For the fallback, consider the event that date t is the first date on which a null entrant is certified. Every earlier date whose level was passed along the run that feeds t had all its entrants certified, and none of them null, so the run consists of non-null dates; the runs that feed different first-failure dates are disjoint, and the levels available to null dates sum to at most α, as in Corollary 2 of Banahene and Opoku [5]. Passing a date's level on when only some of its entrants were certified would break this: a date holding a certified non-null entrant and an uncertified null one would sit inside the run and have its level counted twice. □

Benchmark	Systems	Cheapest so far	Entrants	Dominated	Record claim (pairwise)			Pareto certificate (normal)			Pareto certificate (exact)		
					unadj.	spending	fallback	unadj.	spending	fallback	unadj.	spending	fallback
assistantbench	32	2	1	29	0	0	0	0	0	0	0	0	0
colbench backend programming	22	3	7	12	3	2	2	3	2	2	3	2	2
colbench frontend design	14	3	1	10	0	0	0	0	0	0	0	0	0
corebench hard	60	6	12	42	1	0	0	1	1	1	1	0	0
gaia	36	5	11	20	1	0	0	1	0	0	1	0	0
online mind2web†	32	6	10	16	1	1	1	1	1	1	1	1	1
scicode	37	4	0	33	0	0	0	0	0	0	0	0	0
scienceagentbench	25	3	5	17	0	0	0	0	0	0	0	0	0
swebench verified mini	39	5	10	24	2	1	1	2	1	1	2	1	1
taubench airline	47	5	7	35	1	0	0	1	0	0	1	0	0
usaco	12	1	5	6	3	2	2	4	3	3	3	3	3
All	356	43	69	244	12	6	6	13	8	8	12	7	7

Table 9: Frontier entrants on each HAL board, walking the submissions in date order with each system's mean cost per task fixed on arrival. Cheapest so far: the system was cheaper than everything before it and joined the frontier with no rival to beat (no test). Entrants: the system beat every cheaper system on arrival and was tested against the most accurate of them with the selective record test and the per-date factor. Dominated: a cheaper earlier system was at least as accurate, so the system was never on the frontier. Record claim: the pairwise selective test of Proposition 5; Pareto certificate: the composite test of Proposition 6. For each, certified at the 5% level with only the per-date factor applied (unadj.), under alpha-spending, and under the online fallback, the levels being spent per date with an entrant; spending and fallback control the family-wise error over the board's whole chain.

Indexing each same-date entrant by its own step would make an entrant's level depend on its rank within the date, which is settled only after the date's results are seen; counting dates avoids this and spends one level per date.

Remark 4 (random costs)**.** *When costs are measured per task, $C(b)$ is itself random, and b's mean cost comes from the same run as its outcomes (tokens are spent on failed tasks too), so the holder is not strictly free of b. For certification we fix each system's cost at its mean on the day of its arrival and check the consequence by resampling every system's per-task costs and rebuilding the chain: on the boards with per-task costs, an entrant keeps the same holder in 0.82 to 1.00 of 200 resamples, the number of entrants moves within 50 to 73 around 59, and the number certified under alpha-spending within 5 to 7 around 5. The frontier set of Section 3.3 is where the cost uncertainty is carried in full.*

A.3 Entrants on HAL's own timeline

Table 9 applies the rule of Section A.2 to each board's history. Of 356 arrivals, 43 were the cheapest system so far and joined the frontier without a rival to beat, 244 were dominated on arrival by a cheaper system and were never on it, and 69 (19%) beat every cheaper system and were frontier entrants in the sense a leaderboard would announce. Tested against their most accurate cheaper rival, 12 of the 69 pass at the 5% level with only the per-date factor applied, and 6 under alpha-spending and 6 under the online fallback, the levels being spent over the 55 dates on which an entrant arrived. The composite test of Proposition 6, which certifies Pareto membership rather than a record claim, passes 13 at the 5% level and certifies 8 under alpha-spending and 8 under the fallback; 6 entrants are certified by both rules, 0 by the record claim only and 2 by the Pareto certificate only. The exact randomization form of the certificate (Proposition 7) passes 12 at the 5% level and certifies 7 under alpha-spending and 7 under the fallback. The certified steps, under either certificate, are the large, well-measured ones, 6 to 44 points on boards with at least 45 tasks. The uncertified entrants fall into overlapping groups: 20 beat their rival by under five points on boards of at most 165 tasks, within about one standard error; 47 of the 69 arrived on a date with other submissions, which the per-date factor prices; and 6 pass the 5% level after that factor but not the level left in the alpha-spending budget by the time they arrive, the price of error control over the whole chain. A maintainer who announced every entrant as "new on the frontier" would have made 69 claims, of which 6 to 8 survive a guarantee, depending on the claim and the rule.

				Bias, points (standard error)				Coverage of the 95% interval			
Systems	Tasks	Winners	Pool	reported	bootstrap	iterated	conditional	reported	bootstrap	iterated	conditional
8	50	796	4.6	+1.3 (0.2)	-0.7 (0.2)	-0.8 (0.3)	-0.7 (0.3)	0.95	0.91	0.90	0.96
8	150	841	4.6	+0.8 (0.1)	+0.0 (0.1)	+0.0 (0.1)	-0.2 (0.2)	0.94	0.92	0.90	0.95
8	500	836	4.8	+0.3 (0.1)	+0.0 (0.1)	+0.0 (0.1)	-0.2 (0.1)	0.95	0.94	0.93	0.96
20	50	1242	9.3	+2.2 (0.2)	-0.5 (0.2)	-0.7 (0.2)	-0.4 (0.2)	0.93	0.90	0.89	0.96
20	150	1319	9.2	+1.1 (0.1)	+0.2 (0.1)	+0.2 (0.1)	-0.1 (0.1)	0.94	0.92	0.91	0.95
20	500	1328	9.1	+0.3 (0.1)	-0.0 (0.1)	-0.0 (0.1)	-0.3 (0.1)	0.95	0.93	0.92	0.95

Table 10: The frontier winner's curse on known truth, 300 boards per scenario, every frontier member with at least one cheaper rival. Winners: the number of such members over the 300 boards; pool: mean number of systems that cost no more. Bias: mean of estimate minus true accuracy, in points, with its Monte Carlo standard error. Coverage: share of 95% intervals that contain the true accuracy.

The counts depend on how submissions are batched, and we report that dependence. Batching by calendar week instead of by date gives 56 entrants and certifies 6 as record claims and 7 as Pareto members; reversing the order within a date, worst first instead of best first, gives 100 entrants, 7 and 9. The number of entrants moves because the order decides which system faces which cheaper rival. The number certified stays between 6 and 7 as record claims and between 7 and 9 as Pareto members, across the three rules.

B The frontier winner's curse

A system reported as "on the frontier" was selected for beating every cheaper system, and its reported accuracy is inflated by the selection, like a leaderboard winner's. The selection event, $\bar{y}_b \geq \bar{y}_j$ for all $j \in C(b)$, is a polyhedron in the vector of accuracies, so the conditional method and the iterated bootstrap of Banahene and Opoku [5] apply with the pool $C(b) \cup \{b\}$ in place of the whole board. The conditional estimate is the median-unbiased estimate under the truncated normal law of $\bar{y}_b$ given that it is the largest of its pool [1, 16]. The bootstrap finds the winner of the pool again in every task resample and takes its mean excess over its own observed accuracy as the bias; a second bootstrap level removes the bias the first correction leaves. Costs are fixed at their means, as in certification. A system with no cheaper rival, the cheapest on the board, has a pool of one and nothing to correct.

Table 10 checks both corrections on the simulated boards of Section 5. The reported accuracy of a frontier member overshoots the truth by +0.3 to +2.2 points on average, the larger values on the crowded boards of 20 systems; the iterated bootstrap leaves a bias of at most 0.8 points and the conditional estimate at most 0.7. The reported interval covers 0.93 to 0.95, the conditional interval 0.95 to 0.96, the one-step bootstrap interval 0.90 to 0.94 and the iterated bootstrap interval 0.89 to 0.93. The two bootstrap corrections remove bias at the cost of coverage, as the companion paper found for the leaderboard winner, and only the conditional estimate improves on both. The frontier curse is milder than the leaderboard winner's curse of the companion paper because the pool is smaller than the board, and it is strongest for the expensive systems at the top of the staircase, whose pools are largest. Only 75 to 92% of the frontier members with a rival are truly undominated, which is the plain frontier's error rate seen from the other side.

B.1 On HAL

Table 11 applies the corrections to the 64 systems on HAL's observed frontiers, 53 of which have a cheaper rival. The iterated bootstrap takes 1.7 points off a frontier member's reported accuracy on average and 9.4 at most; the conditional estimate takes 1.1 on average. For 11 of the 64 members, on 5 boards, the adjusted accuracy falls below the reported accuracy of the strongest cheaper rival: at its adjusted accuracy the system would not be on the frontier at all. The corrections are largest where the tasks are few and the frontier members many, CORE-Bench hard and tau-bench airline, and negligible on USACO and ColBench backend, where the tasks are many.

Benchmark	On frontier	With a cheaper rival	Shrinkage, iterated bootstrap (pt) mean	Shrinkage, iterated bootstrap (pt) max	Shrinkage, conditional (pt) mean	Below strongest cheaper rival after adjustment
assistantbench	3	2	3.3	5.3	1.1	1
colbench backend programming	9	8	0.3	0.5	0.1	0
colbench frontend design	4	3	0.2	0.5	0.4	1
corebench hard	8	7	4.3	7.7	3.5	4
gaia	10	9	1.2	2.4	1.3	2
online mind2web†	6	5	0.6	0.9	0.1	0
scicode	3	2	0.2	0.5	0.1	0
scienceagentbench	4	3	0.8	1.6	0.1	0
swebench verified mini	6	5	1.0	2.3	0.1	0
taubench airline	7	6	4.4	9.4	2.8	3
usaco	4	3	0.0	0.1	0.0	0
All	64	53	1.7	9.4	1.1	11

Table 11: Selection-adjusted accuracy of the systems on HAL's observed frontiers. Shrinkage: reported accuracy minus the adjusted estimate, in points, over the frontier members that have a cheaper rival. The last column counts frontier members whose iterated-bootstrap estimate falls below the reported accuracy of their strongest cheaper rival. Every member is listed in the supplement.

Benchmark	Winners	Bias, points reported	Bias, points iterated	Bias, points conditional	Coverage of the 95% interval reported	Coverage of the 95% interval iterated	Coverage of the 95% interval conditional	Placebo coverage
ColBench backend	1465	+0.5	-0.1	-0.3	0.94	0.89	0.94	0.92
ColBench frontend	693	+0.6	+0.2	-0.1	0.87	0.85	0.92	0.91
GAIA	1578	+2.3	+0.2	+0.3	0.88	0.84	0.92	0.89
Online-Mind2Web†	1010	+1.7	-0.1	-0.2	0.92	0.89	0.96	0.92
ScienceAgentBench	682	+3.5	+0.9	+1.6	0.91	0.90	0.94	0.89
USACO	605	+0.1	+0.1	+0.1	0.97	0.97	0.97	0.97

Table 12: The winner's curse on held-out halves of the HAL boards with at least 100 tasks, 200 random splits each. For the frontier members of half A that have a cheaper rival (winners), the bias of the A-half reported accuracy and of the two corrected estimates against the B-half accuracy, shifted by the mean over all systems of the B-half minus A-half accuracy so that the halves' difficulty difference cancels, and the coverage of each 95% interval. Placebo coverage: the coverage of the same reported-accuracy interval for every system on the board, selected or not.

B.2 Held-out halves

In the same splits, the reported accuracy of the frontier winners (6033 winner-splits in all) overshoots the fresh half by +0.1 to +3.5 points on average, depending on the board. The iterated bootstrap is within 0.9 points of it and the conditional estimate within 1.6. Coverage cannot be read against a fresh half the way it can against a known truth, because the half-B accuracy is itself a noisy measurement: a correct interval covers it less often than its nominal level. The placebo column shows how much less, as the coverage of the same reported-accuracy interval for every system on the board, selected or not, at 0.89 to 0.97. The winners' reported intervals cover 0.87 to 0.97, the iterated 0.84 to 0.97 and the conditional 0.92 to 0.97. Coverage therefore does not separate the corrected intervals from the reported ones on these boards, and the bias comparison, which does, is the evidence for the correction.

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